KAROL RAPHAEL S. PAL **BSIT-3**

Capture the Flag 2: Widgets

Goal: Turn the single-card application into a simple multi-section interface.

🏴 Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing `feature/widgets`

```
Administrator@DESKTOP-VKRMBR2 MINGW64 ~/Documents/my_first_flutter_app (feature/widgets)
$ git branch
* feature/widgets

Administrator@DESKTOP-VKRMBR2 MINGW64 ~/Documents/my_first_flutter_app (feature/widgets)
$
```

🏴 Flag 1 – Break It Apart

Objective: Separate your profile information into individual `Text` widgets.

If your card currently looks like this:

Text(

'Chiara Marie C. Canque\n'

'BSIT - 3\n'

'My First Flutter Application\n'

'August 11, 2026',

)

Your task: Separate each piece of information into its own `Text` widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

Text → Application Title

Text → Date

Do not worry about making it pretty yet.

Screenshot:

- Your code showing the separate `Text` widgets
- Your application





```
Text("My First Flutter Application",
style: TextStyle(
  fontSize: 20,
  color:  const Color.fromARGB(255, 10, 70, 119),
  fontWeight: FontWeight.bold,
), // TextStyle
), // Text

Text("August 31, 2026",
style: TextStyle(
  fontSize: 20,
  color:  const Color.fromARGB(255, 10, 70, 119),
  fontWeight: FontWeight.bold,
), // TextStyle
), // Text
```

🏴 Flag 2 – Stack the Information

Objective: Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

Course & Section

Application Title

Date

Screenshot:

- The `Column` in your code
- The four separate pieces of information displayed vertically

```
child: Column(
mainAxisAlignment: MainAxisAlignment.center,
children: [

Image.asset(
  "assets/images/image.png",
  width:100,
  height:100,
), // Image.asset

Text("Karol Raphael S. Pal",
style: TextStyle(
  fontSize: 20,
  color: const Color.fromARGB(255, 10, 70, 119),
  fontWeight: FontWeight.bold,
), // TextStyle
), // Text

Text("BSIT-3",
style: TextStyle(
  fontSize: 20,
  color: const Color.fromARGB(255, 10, 70, 119),
  fontWeight: FontWeight.bold,
```

```
Text("My First Flutter Application",
style: TextStyle(
  fontSize: 20,
  color: const Color.fromARGB(255, 10, 70, 119),
  fontWeight: FontWeight.bold,
), // TextStyle
), // Text

Text("August 31, 2026",
style: TextStyle(
  fontSize: 20,
  color: const Color.fromARGB(255, 10, 70, 119),
  fontWeight: FontWeight.bold,
), // TextStyle
), // Text
```

🏴 Flag 3 – Add Another Element

Objective: Add another individual widget to your `Column`.

Add something that belongs to your profile.

Examples:

Hobby

It must be its **own widget**, not additional text inside another `Text`.

Screenshot:

- New widget in your `Column`
- Updated application

```
Text("Hobby: Sleeping",
 style: TextStyle(
  fontSize: 20,
  color:  const Color.fromARGB(255, 10, 70, 119),
  fontWeight: FontWeight.bold,
 ), // TextStyle
), // Text
```



🏴 Flag 4 – Put Two Things Side-by-Side

Objective: Discover the `Row` widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21 Birthdate: August 11, 2003

Research:

`Row()`

Place the `Row` inside your existing `Column`.

Your structure should now resemble:

Column

├── Text

├── Text

├── Text

├── Text

├── Text

└── Row

    ├── Text

    └── Text

Screenshot:

- Your widget structure/code
- The two items displayed side-by-side



```
            color:  const Color.fromARGB(255, 10, 70, 119),
            fontWeight: FontWeight.bold,
          ), // TextStyle
          ), // Text

          Text("Hobby: Sleeping",
           style: TextStyle(
            fontSize: 20,
            color:  const Color.fromARGB(255, 10, 70, 119),
            fontWeight: FontWeight.bold,
           ), // TextStyle
          ), // Text

          Row(
            children: [
              Text("Age: 21"),
              Text("Birthdate: April 28,2005"),
            ],
          ) // Row
```

🏴 Flag 5 – Give It Some Space

Objective: Improve the spacing between your widgets.

Research:

`SizedBox()`

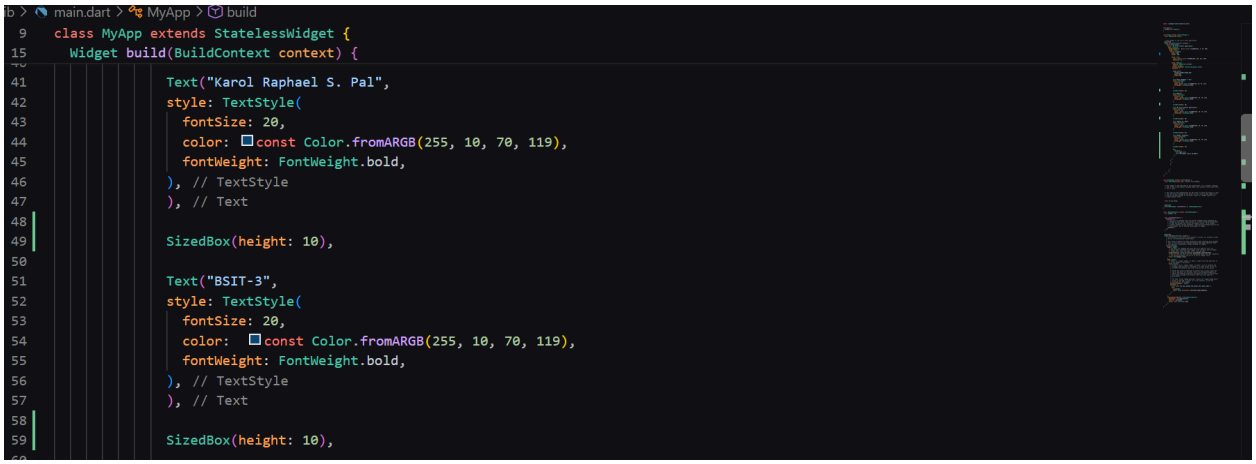
Add appropriate spacing between your:

- Image
- Name
- Course
- Other information
- Row

Challenge: Make the card easier to read **without changing its colors or fonts yet**.

Screenshot:

- `SizedBox` usage
- Updated application



```
Text("My First Flutter Application",
style: TextStyle(
  fontSize: 20,
  color:  const Color.fromARGB(255, 10, 70, 119),
  fontWeight: FontWeight.bold,
), // TextStyle
), // Text

SizedBox(height: 10),

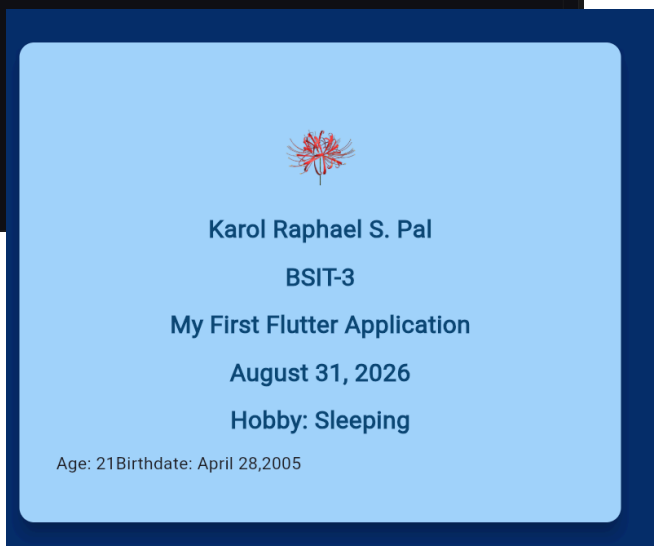
Text("August 31, 2026",
style: TextStyle(
  fontSize: 20,
  color:  const Color.fromARGB(255, 10, 70, 119),
  fontWeight: FontWeight.bold,
), // TextStyle
), // Text

SizedBox(height: 10),
```

```
Text("Hobby: Sleeping",
 style: TextStyle(
  fontSize: 20,
  color:  const Color.fromARGB(255, 10, 70, 119),
  fontWeight: FontWeight.bold,
 ), // TextStyle
), // Text

SizedBox(height: 10),

Row(
  children: [
    Text("Age: 21"),
    Text("Birthdate: April 28,2005"),
  ],
) // Row
```



🏴 Flag 6 – Make It Look Better

Objective: Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

Apply different styling to at least **three** elements.

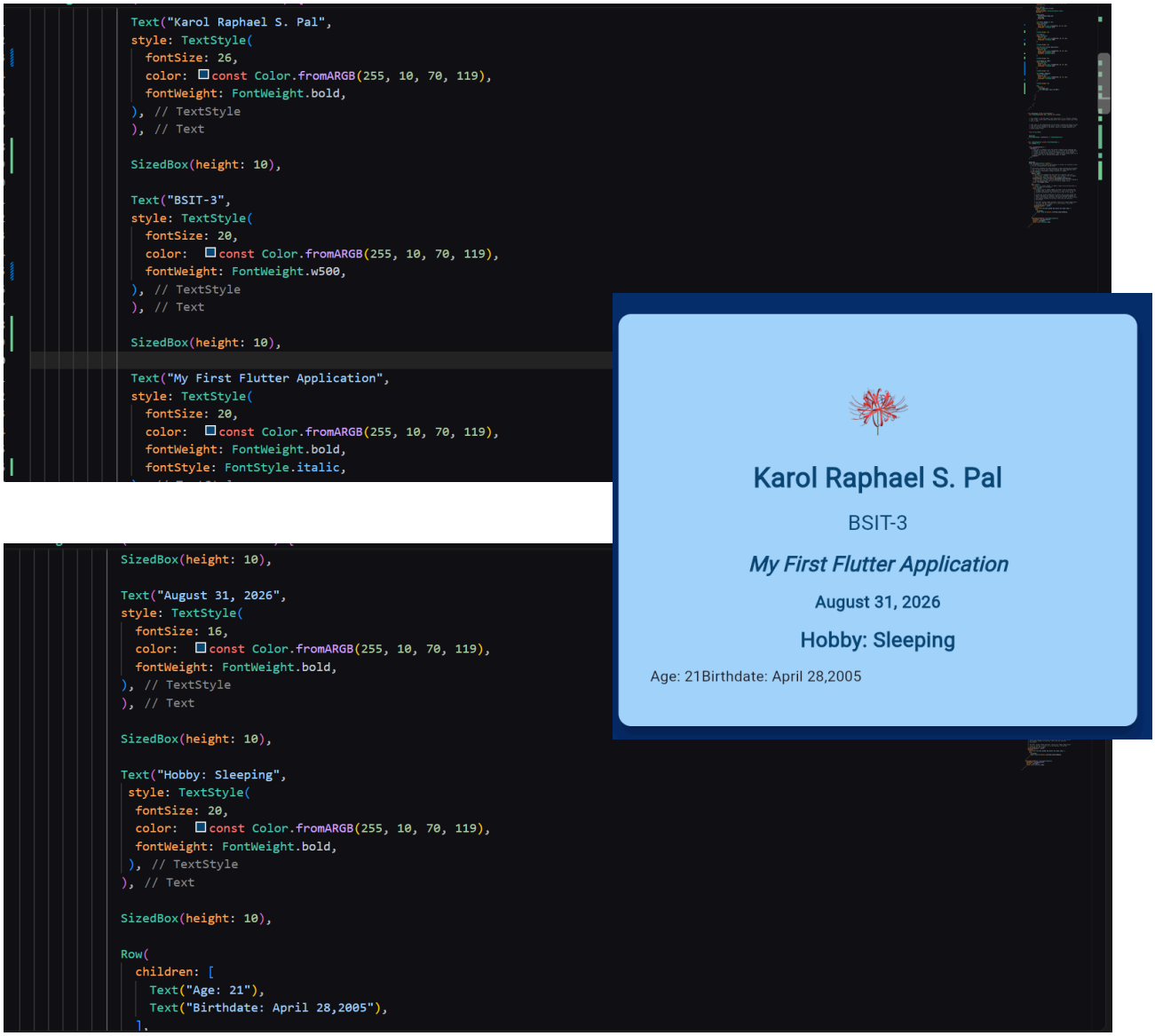
For example:

- Name → larger and bold

- Course → medium size
- Date → smaller
- Application title → italic

Screenshot:

- Your `TextStyle` changes
- Updated application



🏴 Flag 7 – Improve the Layout

Objective: Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

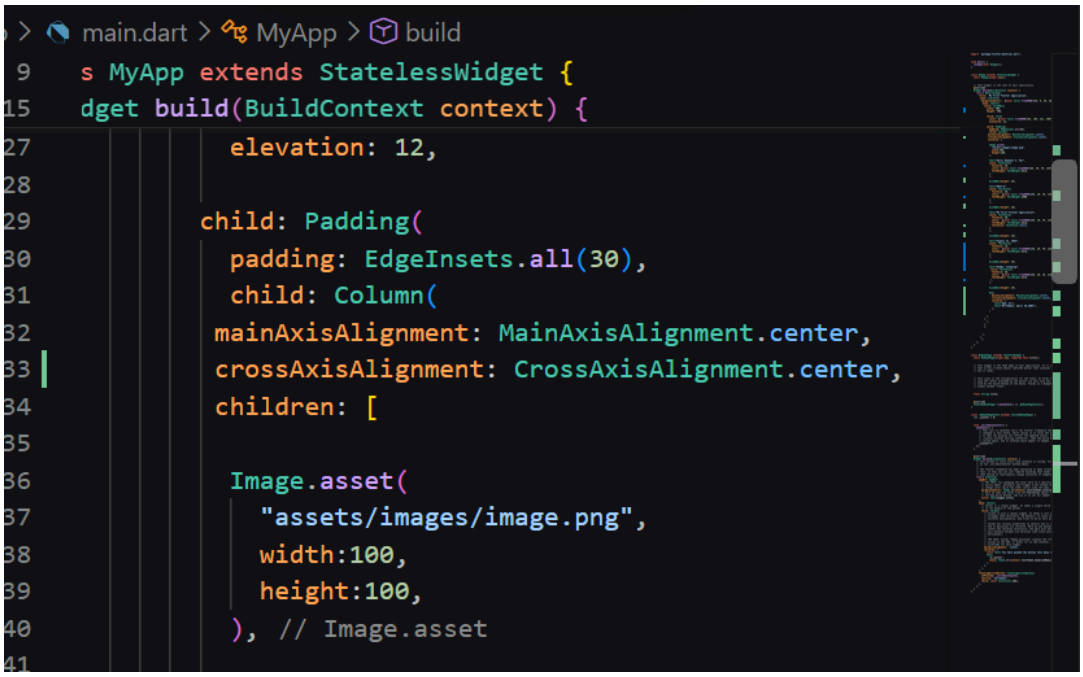
`mainAxisAlignment`

`crossAxisAlignment`

Change the alignment of your content so the card looks intentional rather than simply stacked.

Screenshot:

- The properties you changed
- Updated application



```
Row(
  mainAxisAlignment: MainAxisAlignment.center,
  crossAxisAlignment: CrossAxisAlignment.center,
  children: [
    Text("Age: 21"),
    Text("Birthdate: April 28,2005"),
  ],
```



🏴 Flag 8 – Create a Second Card

Objective: Apply what you learned to create another section.

Create a second `Card` containing something different from your profile.

For example:

My Favorites

Favorite Game

Favorite Food

Favorite Movie

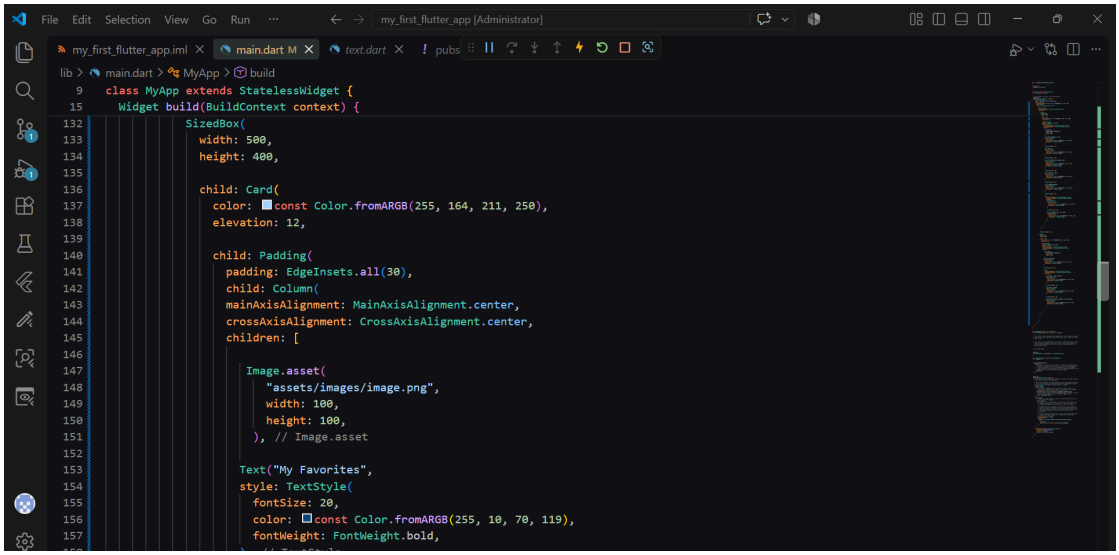
The second card should also use:

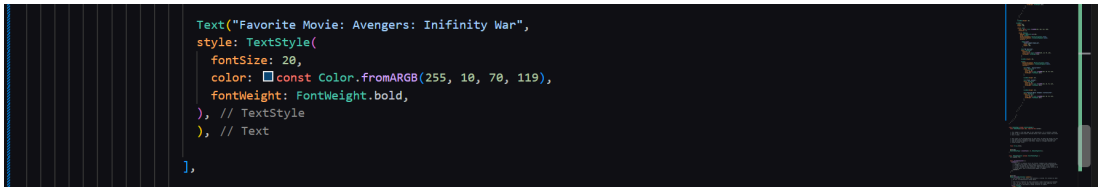
- Individual `Text` widgets
- A `Column`
- At least one `Row`
- Spacing

Screenshot:

- Code for the second card
- Both cards displayed







🏴 Flag 9 – Stack the Cards

Objective: Arrange your two cards vertically.

You now have:

Profile Card

↓

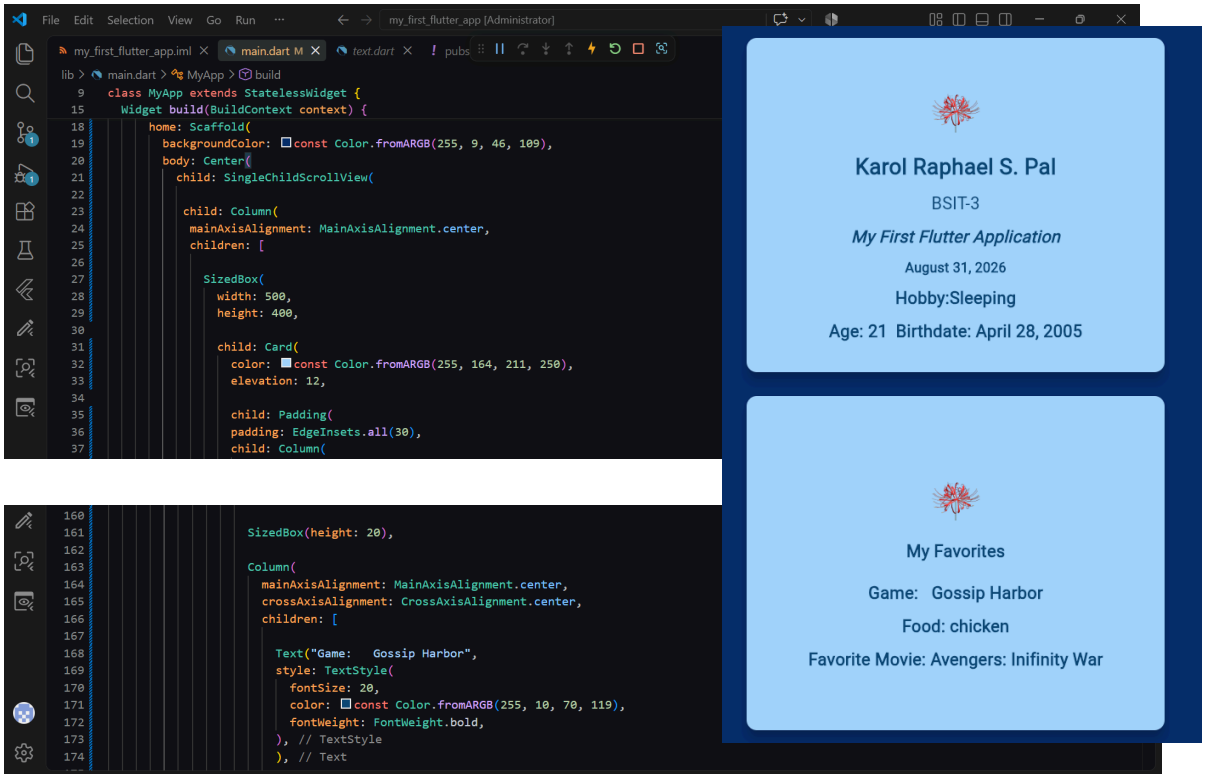
Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

Screenshot:

- The widget containing both Cards
- Both cards visible in the application



🏴 Flag 10 – Widget Discovery

Objective: Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

Examples:

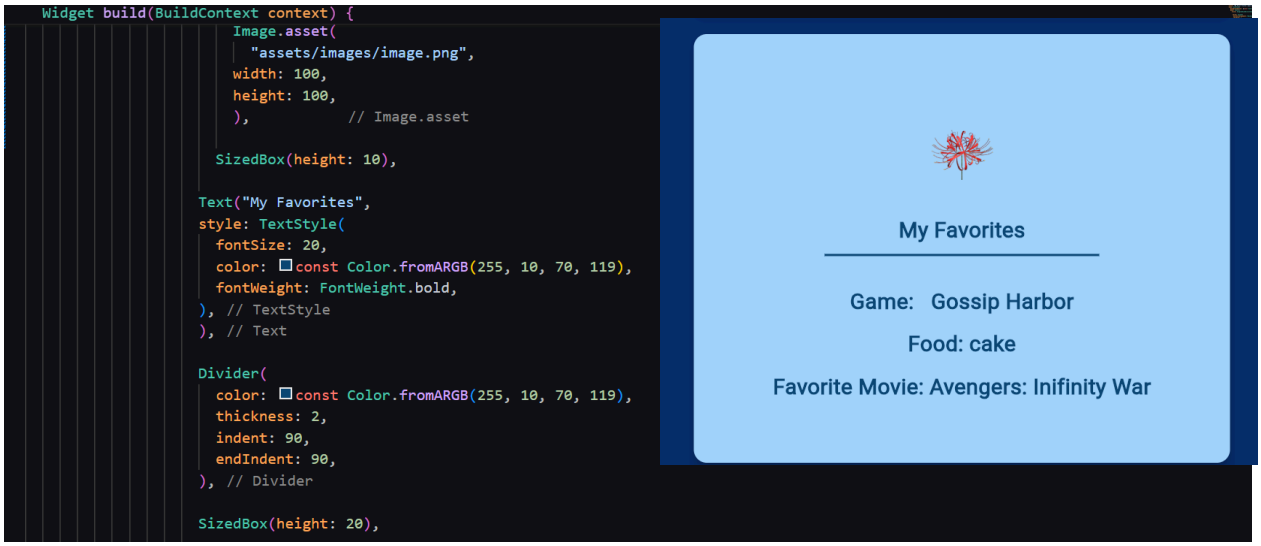
- `Icon`
- `Divider`

- `CircleAvatar`
- `Chip`
- `ElevatedButton`

Add it somewhere appropriate in your application.

Screenshot:

- Your newly discovered widget in the code
- The widget visible in the application



🏴 Flag 11 – Real Phone Test 📱

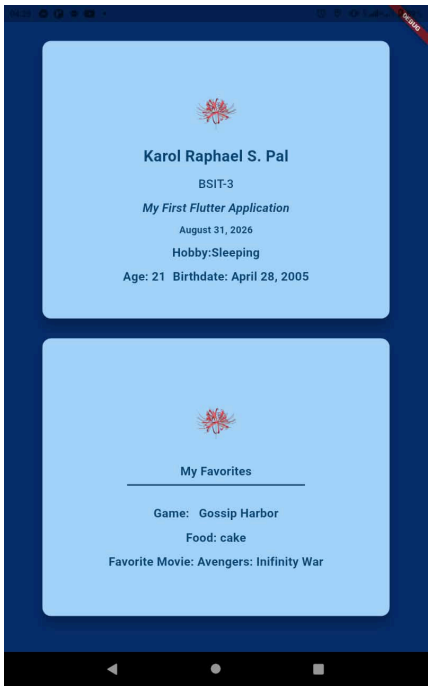
Objective: Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

Screenshot:

- Your completed application running on the physical phone (screenshot from your actual
-

phone)



🏴 Flag 12 – Save Your Progress

Objective: Commit your completed Widget branch.

Follow proper standards of commit messages

Screenshot:

- Successful commit
- `feature/widgets` branch

```
$ git push origin feature/widgets
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 1.27 KiB | 432.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Palkarol/IT314AB_Pal.git
   0492bc4..5ad4128  feature/widgets -> feature/widgets

Administrator@DESKTOP-VKRMBR2 MINGW64 ~/Documents/my_first_flutter_app (feature/widgets)
$
```